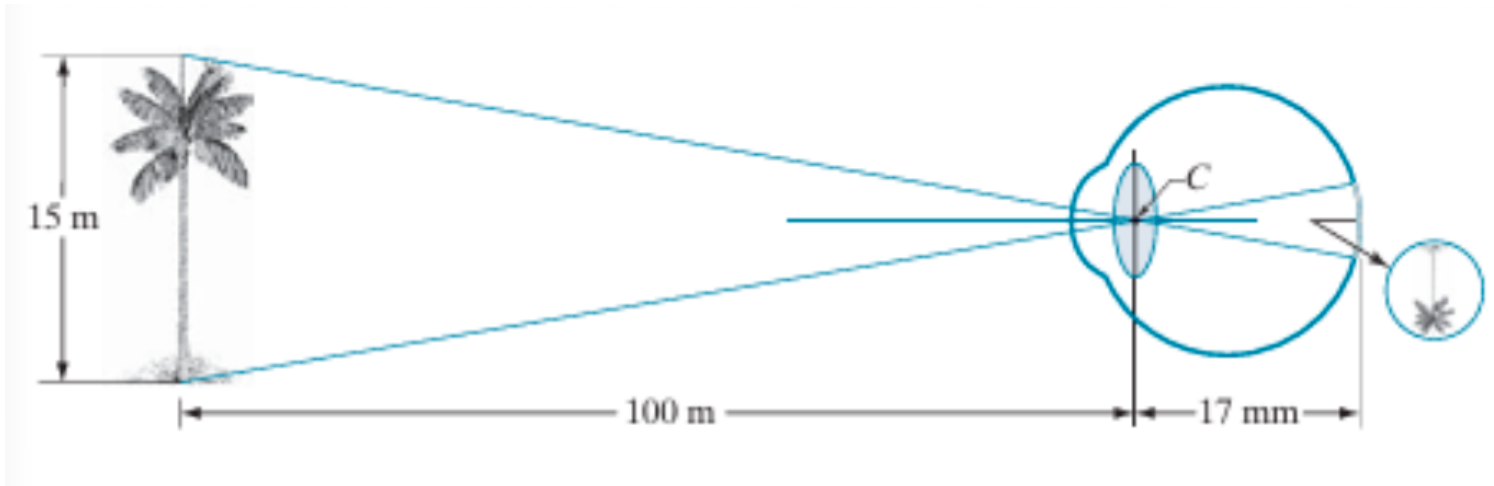


Q3: With neat diagram, describe the image formation by HVS.
With neat sketches, describe briefly brightness adaptation, discrimination and illusion.

- * describe the image formation by HVS
- * describe briefly brightness adaptation, discrimination and illusion

describe the image formation by HVS



Eye lens is flexible as opposed that in camera.
Focal length of eye varies in between 14 mm –17 mm.

Perception :

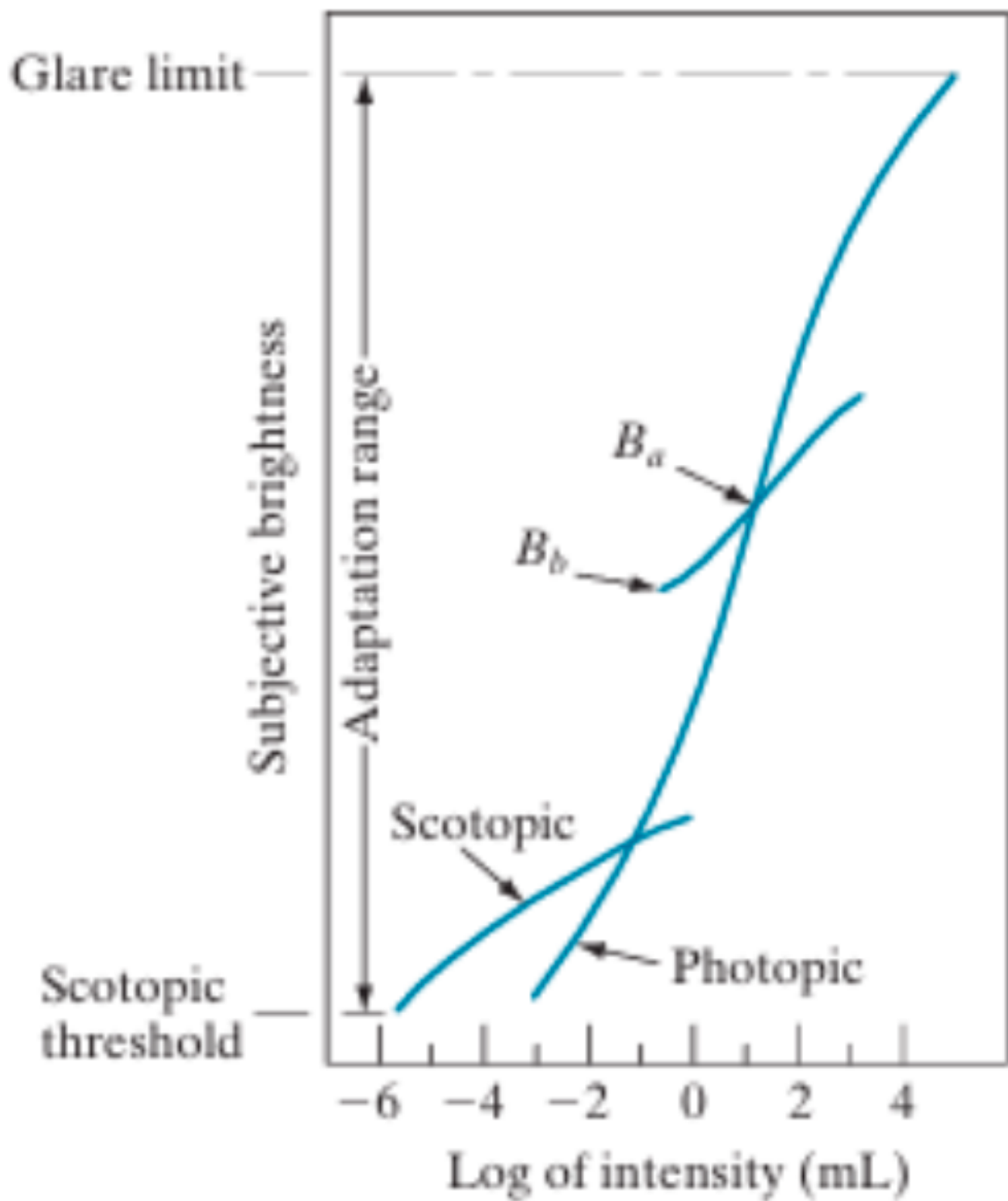
- (i) radiant energy to electrical impulses.
- (ii) decoded impulses by brain.

brightness adaptation

➤ Subjective brightness is a logarithmic function of light intensity

➤ Although adaptation range for photopic vision is quite high, overall adaptation due to scotopic vision is quite low.

➤ Thus, for a given set of brightness condition, the range of discriminative intensity level of HVS is rather small.



Brightness Discrimination refers to the HVS's ability to detect the smallest noticeable change in intensity (ΔI) at a specific background intensity (I).

- **Weber Ratio:** This is represented by the ratio $\Delta I_c / I$, where ΔI_c is the increment of illumination discriminable 50% of the time.
- **Interpretation:** A small Weber ratio indicates good brightness discrimination (the eye can see small percentage changes), while a large ratio indicates poor discrimination.
- **Experiment:** Discrimination is typically measured by showing a subject a uniform background of intensity I and adding a brief flash of intensity ΔI in the center.

Brightness Illusion

Brightness Illusions are phenomena where the perceived brightness of an area differs from its actual physical intensity because the brain's perception is not a simple function of luminance.

- **Simultaneous Contrast:** A classic example where the perceived brightness of a region depends on its background. A gray square appears lighter when placed on a black background than when placed on a white background, even though the squares are physically identical.
- **Mach Bands:** The appearance of illusory bright and dark bars at the transition points of an intensity ramp, caused by lateral inhibition in the retina.